



Digital Campus Noticeboard & Communication Portal

ANANTHARAMAN P.K

MGP21NMCA012

S10 IMCA, Department of Computer Applications,
Saintgits College of Engineering (Autonomous), Pathamuttom -686532

Under the Guidance of

Er. Vidya S

ASSISTANT PROFESSOR

Department of Computer Applications,
Saintgits College of Engineering (Autonomous), Pathamuttom -686532

18-03-2026

Review 3 Presentation

1



Contents

1	Introduction
2	Existing Applications
3	Objective
4	Target Audience
5	Technologies used
6	Database Details
7	System Architecture
8	Development Process
9	Features and Functionalities
10	Technical Challenges
11	Testing Methods
12	Future Enhancements
13	Conclusion

18-03-2026

Review 3 Presentation

2



Introduction

- The Digital Campus Noticeboard is an intelligent, centralized communication platform designed to replace chaotic WhatsApp groups and ignored physical bulletin boards.
- Features a strict hierarchy where Staff can draft notices, Admins approve/reject them, and Students view/acknowledge them based on their specific department (MCA, BTech, MBA).
- Utilizes background schedulers and Django signals to automatically fire off targeted HTML emails, in-app notifications, and deadline warnings.
- Built using the Django Framework, it guarantees high security, rapid development, and provides administration with real-time analytics on student engagement..

18-03-2026

Review 3 Presentation

3



Existing Applications

Traditional Physical Noticeboards

- Highly localized and easily ignored by students.
- Zero accountability: No way to verify if a specific student actually read a crucial deadline.

WhatsApp Groups / Generic Emails

- Highly cluttered; important academic notices get lost in informal chatter.
- Lacks a structured approval process.
- No built-in attendance tracking or analytics for the administration to measure engagement..

18-03-2026

Review 3 Presentation

4



Objective

- To centralize campus communication by providing a single source of truth for all verified university announcements and deadlines.
- To enforce accountability using a "Mark as Read" acknowledgment system, allowing admins to track exactly who has and hasn't seen a notice.
- To automate the communication lifecycle, using background tasks to send 48-hour and 24-hour expiration warnings to unread students..
- To streamline administrative workflows by giving management a dedicated analytics dashboard to monitor department-wise engagement and poll responses.

18-03-2026

Review 3 Presentation

5



Target Audience

- **Students (MCA, BTech, MBA):** Need a clean portal to check specific department news, integrate deadlines and vote on campus polls.
- **Staff :** Need an intuitive platform to draft notices, attach files, and receive instant feedback (approval/rejection) from management.
- **System Admin :** Admin who has full control over what gets published, and require analytical data on student compliance.

18-03-2026

Review 3 Presentation

6



Technologies Used

➤ Frontend:

- HTML5, CSS3, Bootstrap 4: For responsive, "glass-morphism" styled UI components.
- JavaScript & jQuery: For dynamic elements like live-search and asynchronous AJAX requests.

➤ Backend:

- Python & Django : Handles core logic, user authentication.
- Django Signals: Triggers automated events.

➤ Database Used:

- SQLite: Lightweight, file-based relational database management used for rapid development and local testing.

18-03-2026

Review 3 Presentation

7



Database

1. Notice & Profile Models

- **Notice:** Stores title, message, target_department, status (pending/approved/rejected), and tracks automated alerts.
- **Profile:** Extends the default User model to assign students to specific departments (MCA, BTech, MBA) for targeted delivery.

2. Engagement Models

- **NoticeReadStatus:** A relational table linking a User to a Notice, acting as a digital "Read Receipt" timestamp.
- **Poll & PollResponse:** Stores active campus questions and tracks unique 'Yes/No' responses.

18-03-2026

Review 3 Presentation

8



Database

Users Table

id	password	last_login	is_superuser	username	last_name	email
1	pbkdf2_sha256\$12000005\$HvYAKDThfa...	2026-03-13 05:49:16.905261	1	AnantharamanPR		anantharamann003@gmail.com
2	pbkdf2_sha256\$12000005\$4MSD9grnVIB8...	2026-03-13 05:49:31.827441	0	WalterWhite	White	
3	pbkdf2_sha256\$12000005\$5eetFeH2h43eb...	2026-03-05 05:56:47	0	Kevin		kkt37bro@gmail.com
4	pbkdf2_sha256\$12000005\$7bPv0194cU...	2026-03-12 11:11:11.731243	0	Adwaith		adwaithkrishna178@gmail.com
5	pbkdf2_sha256\$12000005\$XicXqcd79yT...	2026-02-17 07:06:09.556463	0	Daniel		
6	pbkdf2_sha256\$12000005\$9qskuv6AcpA1F...	2026-02-17 07:06:43.357128	0	Adam		
7	pbkdf2_sha256\$12000005\$9c9uCGmGt8F8...	2026-02-17 07:07:29.196005	0	Chris		
8	pbkdf2_sha256\$12000005\$CPEYIOXA7c0OC...	2026-02-17 07:08:31.139297	0	HarveySpecter	Specter	
9	pbkdf2_sha256\$12000005\$FurcSgmsBdrSg...	2026-02-17 07:00:16.884606	0	AbrahamPhillip	Phillip	
10	pbkdf2_sha256\$12000005\$bKXlEpAncE8KJO...	2026-02-17 07:07:15	0	John		anantharamanp004@gmail.com
11	pbkdf2_sha256\$12000005\$4acP9yee73QW...	NULL	0	yaseen		anantharamann003@gmail.com

Notices Table

id	message	title	created_at	created_by_id	tags
1	To be held on March 26	Annual Day	2026-02-16 05:51:03.178477	2	General
2	To be conducted on April 17	Sports day	2026-02-16 08:14:23.720386	2	General
3	To be conducted on June 23	Re-Examination Of Second Internal	2026-02-16 08:20:26.553725	2	General
4	Get ready for 'Aara 2026'! Audition...	Annual Cultural Fest 2026	2026-02-17 06:56:08.949341	2	Event
5	All students must submit their ...	Health Insurance Update	2026-02-17 06:56:56.280567	2	Important
6	Due to upcoming mid-term exams, the...	Library Extended Hours	2026-02-17 06:57:33.452921	9	General
7	Mr. Rajesh Khanna, CEO of FinLink, ...	Corporate Leadership Talk	2026-02-17 06:58:28.325006	8	General
8	Register for the Inter-College ...	Case Study Competition	2026-02-17 06:59:13.374409	8	General
9	Join the Robotics Club orientation...	Robotics Club Orientation	2026-02-17 07:01:05.174180	9	General
10	Tech Corp is visiting for Beech CS ...	Campus Recruitment: Tech Corp	2026-02-17 07:01:43.217471	9	Important
11	A hands-on workshop on Spring Boot ...	Java Full Stack Workshop	2026-02-17 07:02:25.937739	2	General

17-03-2026

Review 3 Presentation

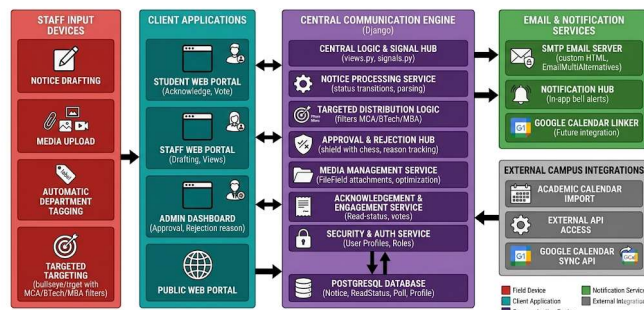
9



System Architecture

- The Template (Frontend): Manages the UI. Uses Django Template Language to dynamically render targeted notices, hide Admin analytics from Staff, and display real-time notification bells.
- The View (Logic): Python controllers that intercept web requests. They filter notices so students *only* see their department, enforce security.
- The Model (Data Layer): Communicates securely with the relational database using the Django ORM to query read-receipts and save poll responses

DIGITAL CAMPUS NOTICEBOARD & COMMUNICATION PORTAL



18-03-2026

Review 3 Presentation

10



System Architecture

1. Frontend

Role: The web-based browser portals where Students, Staff, and Admins visually interact with the platform.

Technologies Used: HTML5, CSS3, Bootstrap 4, and JavaScript/jQuery.

Key Functions :

- Renders customized dashboards based on who logs in (Admin Analytics, Staff Drafting Tools, or Student Department Notices).
- Features responsive, "glass-morphism" UI cards and dynamic color-coded alerts.
- Utilizes asynchronous AJAX (via jQuery) to power the Live Search bar, allowing students to find notices instantly without reloading the page.

18-03-2026

Review 3 Presentation

11



System Architecture

2. Backend

Role: The secure "brain" of the application that enforces rules, routes data, and automates background tasks.

Technologies Used: Python and the Django Web Framework .

Key Functions :

- Enforces strict role-based access control (e.g., ensuring Staff cannot secretly approve their own pending notices).
- Contains the routing logic to ensure an MCA student never accidentally sees a BTech-only notice.
- Uses Django Signals to act as an automation engine—instantly dispatching SMTP HTML emails and in-app bell notifications the exact second an Admin clicks "Approve."

18-03-2026

Review 3 Presentation

12



System Architecture

3. Database

Role: The permanent, organized digital vault that securely records all campus communication and user engagement metrics.

Technologies Used: SQLite

Key Functions :

- Stores relational data such as User Profiles, Department assignments, and the status of Notices (Pending, Approved, Rejected).
- Acts as the accountability tracker by recording timestamped "Read Receipts" (NoticeReadStatus) when a student acknowledges a deadline.
- Securely logs unique Poll Responses to enforce a strict "one student, one vote" policy.

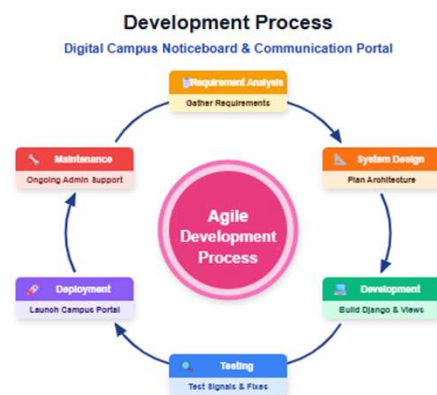
18-03-2026

Review 3 Presentation

13



Development Process



18-03-2026

Review 3 Presentation

14



Development Process

1. Requirement Analysis

- **Approval Workflow Design:** Identified the necessity of a "Shield" layer where all staff submissions remain in a pending state until verified by an Admin, preventing unregulated spam and ensuring content quality.
- **Targeted Audience Logic:** Mapped out the data filtering requirements to ensure notices are delivered only to relevant student groups (e.g., MCA, BTech, or MBA), reducing information overload for the student body.
- **Accountability Features:** Defined the requirement for digital "Read Receipts" and unique poll IDs to track student engagement and mathematically enforce a "one student, one vote" policy.

2. System Design

- **Database Architecting:** Designed a relational schema to link users, departments, and notices, ensuring data integrity for read receipts and poll results.
- **UI State Mapping:** Planned distinct user journeys for Admins, Staff, and Students to ensure the interface adapts to specific permissions.
- **Component Blueprints:** Created wireframes for the dashboard to organize notices, analytics, and drafting tools into a clean, mobile-responsive layout.

18-03-2026

Review 3 Presentation

15



Development Process

3. Implementation

- **Django Core Development:** Built the robust backend logic using Python/Django to handle notice processing, user authentication, and role-based routing.
- **Frontend Integration:** Developed the client-side interface using Bootstrap 4 to provide a professional, responsive look across all campus devices.
- **Signal Automation:** Wired up Django signals to automate the delivery of customized HTML emails and in-app alerts the moment a notice is approved.

4. Testing & Deployment

- **Security Permission Locks:** Rigorously tested the backend to ensure Staff members absolutely cannot bypass the Admin approval process or edit others' posts.
- **Functional Bug Squashing:** Conducted end-to-end tests of the notice lifecycle—from drafting and approval to student acknowledgment and data logging.
- **Production Configuration:** Prepared the environment for deployment, transitioning from local SQLite testing to a scalable PostgreSQL setup for live campus use.

18-03-2026

Review 3 Presentation

16



Development Process

5. Maintenance

- **Performance Monitoring:** Conducted regular checks to ensure the portal remains responsive as the volume of archived notices and student logs increases.
- **System Refinement:** Provided ongoing updates to the analytics engine to improve the accuracy of department-wise engagement reports.
- **Admin Support & Security:** Established a routine for monitoring SMTP logs and security permissions to ensure continuous, reliable delivery of urgent campus notifications

18-03-2026

Review 3 Presentation

17



Features and Functionalities

1. Approval & Rejection Workflow

- **Two-Step Verification:** Staff submissions are held in a "Pending" state until verified by an Admin.
- **Feedback Loop:** Rejections trigger instant alerts with "Admin Notes" explaining why.
- **Correction Path:** Features a seamless "Edit & Resubmit" workflow for rejected content.

2. Automated Communication Engine

- **Real-time Alerts:** Uses Django Signals to trigger in-app bell notifications immediately upon approval.
- **Smart Dispatch:** Sends rich HTML emails via SMTP with direct "Call-to-Action" buttons.
- **Automation:** Eliminates manual announcements by linking database saves to notification delivery.

18-03-2026

Review 3 Presentation

18



Features and Functionalities

3. Analytics & Engagement Tracking

- **Admin Dashboard:** A restricted view displaying real-time data on campus communication.
- **Poll Management:** Tallies live votes and enforces a "one student, one vote" policy for data integrity.

4. Targeted Communication & Filtering

- **Granular Audience Targeting:** Allows Staff to broadcast notices to specific departments (MCA, BTech, MBA), individual years, or even specific user roles.
- **Tagging System:** Notices are automatically tagged with metadata (e.g., "Urgent," "Exam," "Placement") for faster filtering and high visibility.
- **Search & Archive:** Provides a searchable repository of all past notices, ensuring students can find historical information without scrolling through chat history

18-03-2026

Review 3 Presentation

19



Features and Functionalities

5. Admin & Security Controls

- **Role-Based Access (RBAC):** Strict permission levels ensure that only verified Staff can draft notices and only System Admins can finalize them.
- **Expiry Management:** Staff can set "auto-archive" dates; notices automatically disappear from the student feed once the deadline has passed.
- **Audit Logging:** Tracks the history of every notice, showing who created it, who approved it, and any rejection reasons provided.

6. Interaction & Feedback Systems

- **"Mark as Read" Accountability:** A digital read-receipt system that allows Admins to see exactly which students have acknowledged an announcement.
- **Real-time Campus Polls:** Integrated voting system for quick feedback on campus events, with the database enforcing a "one-vote-per-student" policy.
- **Document & Media Attachments:** Supports secure uploads of PDF timetables, JPG event posters, and document links directly within the notice card.

17-03-2026

Review 3 Presentation

20



Technical Challenges

- **Challenge 1: Database Locking (SQLite)**

- **Issue:** Background schedulers checking for deadlines would occasionally clash with a user saving a notice simultaneously, causing a "database is locked" crash.
- **Solution:** Configured the database connection with an explicit timeout option ('timeout': 20), allowing concurrent read/writes to queue gracefully without crashing the server.

18-03-2026

Review 3 Presentation

21



Technical Challenges

- **Challenge 2: View Security and State Leaks**

- **Issue:** Staff members could potentially access the Admin Analytics dashboard by manually typing the URL.
- **Solution:** Implemented strict backend view decorator alongside frontend template logic to physically lock the routes.

18-03-2026

Review 3 Presentation

22



Testing Methods

- **Role-Based Routing Tests:** Logged in simultaneously as Student, Staff, and Admin to ensure students couldn't access creation forms, and staff couldn't see analytical data.
- **Logic Auditing:** Tested the "Mark all as Read" feature to ensure it only acknowledged notices specific to the student's department, ignoring cross-department data.
- **UI/UX Visual Testing:** Ensures a professional and seamless "Campus Portal" experience by verifying consistent branding, implementing helpful "empty state" guidance for pages and auditing navigation elements like breadcrumbs and back buttons to maintain an intuitive user path.

18-03-2026

Review 3 Presentation

23



Future Enhancements

- **PostgreSQL Migration:** Transitioning the database from SQLite to PostgreSQL for the production launch to handle thousands of concurrent student read/writes.
- **Mobile Push Notifications:** Integrating Firebase Cloud Messaging (FCM) to send push notifications directly to student smartphones, bypassing the need for email entirely.
- **Advanced Analytics Export:** Allowing administrators to export "Unread Student" lists as automated PDF or Excel reports for disciplinary tracking.

18-03-2026

Review 3 Presentation

24



Conclusion

- This project successfully modernizes campus communication by bridging the gap between administration and the student body.
- By implementing a strict approval workflow, automated deadline tracking, and targeted department filtering, the system eliminates the noise and spam of traditional messaging apps.
- The inclusion of read-receipts and engagement analytics transforms basic announcements into a highly measurable, proactive, and accountable ecosystem.

18-03-2026

Review 3 Presentation

25



Present Status of the Project

Activities	0th Review	Review 1	Review 2	Review 3
Topic Selection & Approval	Completed	Completed	Completed	Completed
Existing Applications Study	Doing	Completed	Completed	Completed
Technologies Selection	-	Completed	Completed	Completed
Database Creation	-	Completed	Completed	Completed
Module Identification	-	Completed	Completed	Completed
UI Creation	-	Doing	Completed	Completed
Testing Methods	-	-	-	Doing

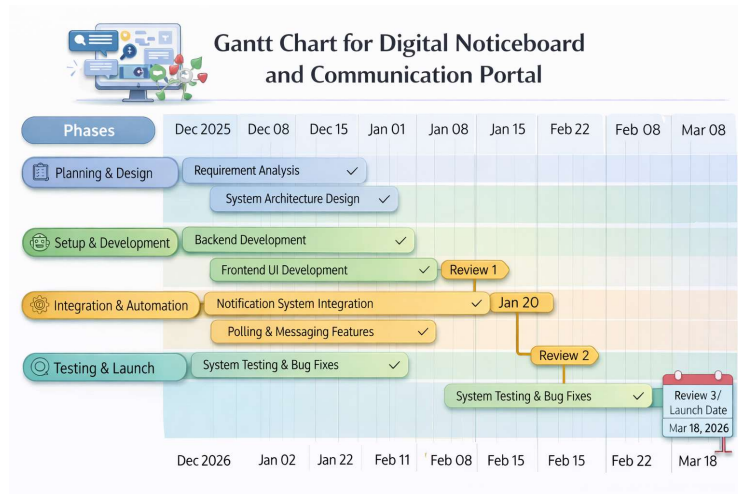
18-03-2026

Review 3 Presentation

26



Gantt Chart



18-03-2026

Review 3 Presentation

27



Thank
you



18-03-2026

Review 3 Presentation

28